

Boundary Compatibility, Not Acyclicity: Structural Penalties for Learning an Edge-to-Cycle Lifting

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Abstract

Moving data between structured representations — graphs, cell complexes, sheaves — can lose information, and homological algebra supplies exact tools for measuring what a map destroys. It is natural to hope that related constraints can also *improve* a learned transformation when used as training objectives. We study two different tasks: selection of a genuine degree-preserving chain map on an annulus, and a primary continuous edge-to-cycle-coordinate lifting. The latter fits $W: R^E \rightarrow R^F$; its transpose is a candidate degree-2 differential, not a chain map between complexes. We test a boundary-of-boundary penalty and two motivated surrogates. Their behavior differs, and cheap analytic diagnostics illuminate the observed differences.

In a prospectively locked same-generator-family replication — with a disclosed sum-versus-mean implementation deviation in the compatibility penalty — on 29 eligible generator seeds, each a joint topology/data/noise realization, we fit one separate map per topology. The hypotheses, directions, endpoint, training size, and weights were informed by earlier work on the same family, whose seed identities were not retained; this is not an independent confirmation or pristine preregistration. A **boundary-compatibility** penalty improves held-out recovery of the learned lifting: Bonferroni-adjusted interval $[-2.749, -1.511]$ on the paired $\log_{10}$ ratio. Paired responses encode B_2 ; the structural penalty itself does not directly use B_2 or the responses, but uses B_1 , which determines the target cycle subspace and is strong side information that the unpenalized baseline ignores. In a prespecified secondary analysis, **defect covaries with held-out error along the compatibility-penalty path**: the mean within-seed Pearson correlation is $+0.961$, with an unadjusted 95% interval $[+0.935, +0.987]$, and is positive in 29 of 29 eligible seeds. The common penalty weight drives both quantities, and the nine fits within a topology are not independent observations; this path result does not establish independent predictive information or off-path calibration.

The same campaign rejects one surrogate objective and finds no improvement from another. A **singular-value cone surrogate**, $\exp(-2 \cdot \sigma_{\min}(W))$, does not merely fail to help; it harms the lifting, adjusted interval $[+0.109, +0.270]$. For an **RTD-inspired normalized pairwise-distance surrogate**, the adjusted interval $[-0.002, +0.038]$ includes zero. The latter asks an intentionally lossy lifting to preserve full source geometry. These are not mapping-cone homology and the published RTD/SRTD statistic, respectively.

We identify two distinct mechanisms. In a second setting — selection among a fixed family of twelve maps on a cellular annulus — every **hard decoded map** is a chain isomorphism and therefore a quasi-isomorphism. Exact cone acyclicity cannot distinguish those twelve vertices: the cone-only model identifies at chance while all 6,000 hard decoded predictions have acyclic cones. Its differentiable soft-mixture objective can vary during training, so this is a limitation of the hard certificate, not a proof that the optimization signal is constant. In the continuous-lifting setting the singular-value surrogate

monotonically rewards inflating the smallest singular value beyond its ground-truth value and imposed a harmful bias here. These mechanisms were identified retrospectively during the post-campaign audit. They require no fitting and are proposed as screens for future objectives, but they were not prospective checks for the reported campaigns and are neither necessary nor sufficient conditions for improved finite-sample prediction.

We do not claim that homological structure teaches a model. We claim that one specific penalty, derivable from the input, measurably improves an edge-to-cycle lifting in this same-family replication. Along its prespecified weight path, compatibility defect covaries with held-out error; the two tested surrogates do not improve this task.

1. Introduction

1.1 The question

Two mathematically different learning tasks must be separated. A conversion between chain complexes is a **typed map**: one degree-preserving component per degree, satisfying the chain-map equations. The annulus experiment selects such a map from a finite family. Homological algebra measures what a chain map destroys (the kernel of its induced homology map), cannot reach (the cokernel), and whether it is a quasi-isomorphism (equivalently, whether its mapping cone is acyclic).

The primary continuous experiment is narrower. It learns $W: R^E \rightarrow R^F$, from edge signals to coordinates in a chosen cycle basis; $W^T: R^F \rightarrow R^E$ is a candidate degree-2 differential. This degree-changing lifting is not itself a typed chain map or a conversion between two complexes. The practical question is whether structural penalties on its implied complex improve recovery of that lifting.

1.2 What is and is not new

Learning higher-order structure from graphs is established: Differentiable Cell Complex Module [5] learns cell probabilities jointly with a task, Differentiable Lifting [6] learns liftings into cellular, simplicial, and combinatorial complexes, and Neural Sheaf Diffusion [7] and Knowledge Sheaves [8] learn sheaf restriction maps. Comparing paired representations through cross-barcode is likewise established: Representation Topology Divergence [1], its differentiable autoencoder form [2], and the symmetric variant SRTD [3] are the direct basis for our divergence reference, and a quotient-homology account of neural representation [4] develops an adjacent algebraic view. Categorical deep learning [9] and functor learning by gradient descent [10] ask what algebraic structure a learned map should respect.

Our contribution is narrow and largely negative-with-a-mechanism:

1. A prospectively specified analysis in a **same-generator-family replication** showing that an input-derivable **boundary-compatibility penalty** improves an edge-to-cycle lifting under scarce paired data, subject to the disclosed protocol implementation deviation and outcome-informed design.
2. A prespecified secondary result, reported with an unadjusted interval, that the **compatibility defect covaries with held-out damage along the fixed compatibility-penalty path**, not that it independently predicts damage.
3. Multiplicity-controlled primary results showing harm from a **singular-value cone surrogate** and no detected improvement from an **RTD-inspired normalized pairwise-distance surrogate**.
4. Two **retrospective no-fit diagnostic heuristics**, proposed as prospective screens for future work, for detecting constant or target-misaligned signals, and a benchmark generator in which homological defects genuinely vary.

This is not a new theorem or algorithm for equality-constrained least squares or Hodge decomposition. Its narrow contribution is an evidence-traceable evaluation of an input-derived cycle-subspace penalty on one deterministic synthetic lifting family. We make no claim of priority over any cited system, and §3 is positioning rather than systematic review.

2. Background and definitions

Each concept is stated in plain language before notation.

2.1 Graphs, cell complexes, and cellular sheaves

A **graph** records which pairs of objects are related. A **cell complex** adds higher-dimensional pieces: on top of vertices (degree 0) and edges (degree 1) it attaches faces (degree 2) glued along closed edge loops. A **cellular sheaf** attaches a small vector space to each cell and a linear transport map to each incidence, so data can move along the complex.

Formally a two-dimensional complex has boundary operators d_1 (edges to vertices) and d_2 (faces to edges) satisfying $d_2 \circ d_1 = 0$.

2.2 Typed maps and chain-map compatibility

A **typed map** carries degree- n data to degree- n data. It is a **chain map** when it commutes with the boundaries — informally, when it does not tear the complex. For a source with boundary d_C and target with boundary d_D ,

$$d_D \circ F_1 - F_0 \circ d_C = 0.$$

The left-hand side is the **chain-map defect**. Our historically named `ExactChainMapLayer` parameterizes (F_0, F_1) inside the nullspace of that expression, so the chain-map equation holds exactly rather than being penalized. This use of “exactly” means equality to numerical tolerance; it does **not** mean that a sequence is exact in the algebraic sense $\text{im } d_{n+1} = \ker d_n$. That utility layer is not used in either campaign reported here: the annulus model mixes a fixed basis of exact chain maps, while the continuous lifting uses a soft penalty.

2.3 The implied complex

This paper's central move. When a model learns a lift W from edge signals to cycle-basis coordinates, W^T is a candidate face boundary. The learned lifting therefore **implies a complex** with boundaries (B_1, W^T) , and that complex is legitimate only if $B_1 \circ W^T = 0$. Every structural term below is written as a condition on the implied complex rather than bolted on:

campaign key	form	meaning
exact	$\text{mean}((B_1 \circ W^T)^2)$	boundary-of-boundary compatibility: the implied operators satisfy $d \circ d = 0$
cone	$\exp(-2 \cdot \sigma_{\min}(W))$	singular-value cone surrogate that rewards increasing the smallest singular value
rtd	MSE between normalized pairwise distances	RTD-inspired distance-preservation surrogate

All three short keys are retained because they were frozen in the campaign protocol and machine-readable result. They must not be read as identities: `exact` is **not exactness of a sequence** — `W = 0` already gives zero loss without implying $\text{im } W^\top = \ker B_1$; `cone` is **not mapping-cone homology**; and `rtd` is **not the published RTD or SRTD statistic** described in §2.5. Exact mapping-cone homology and the corrected RTD/SRTD implementation appear only as separate diagnostics elsewhere in this work.

2.4 Kernels, cokernels, and mapping cones

The **kernel** of the induced map on homology is what a conversion destroys; the **cokernel** is what it cannot reach. A **mapping cone** is acyclic exactly when the map is a quasi-isomorphism, and its homology is the sum of the two:

$$\dim H_n(\text{cone } F) = \dim \text{coker } H_n(F) + \dim \ker H_{n-1}(F).$$

We verify this identity numerically on 24 maps: the twelve dihedral candidates of §4.1 and twelve cycle-killing regression fixtures. It matters here because it shows the cone **bundles** two directional facts into one number and loses their separation — the kernel and cokernel are strictly more informative than the acyclicity bit.

2.5 RTD and SRTD

Representation Topology Divergence compares two point clouds describing the same items by tracking when topological features appear and disappear as a distance threshold grows. **SRTD** is a symmetric variant. Our corrected reference normalizes each dissimilarity matrix by its full-matrix 0.9 quantile, reports persistence separately by degree, uses degree 1 for the published scalar, and caps exact enumeration at 64 entities.

3. Related work

§1.2 and §2.5 cite the primary sources. In summary: RTD [1], RTD-AE [2], and SRTD [3] define the divergence reference, with quotient homology [4] an adjacent algebraic treatment; DCM [5] and DiffLift [6] are the closest precedents for learning higher-order topological structure from graphs; Neural Sheaf Diffusion [7] and Knowledge Sheaves [8] for learned sheaf transports; DeepMoE [11], GMoE [12], GraphMETRO [13], and MvCGE [14] for per-example expert routing.

Categorical deep learning [9] argues architectures are usefully described as algebras over a theory, and functor learning [10] gives an instance where the learned object must preserve structure rather than merely fit data. Our annulus model forms a softmax convex mixture of twelve fixed chain-map templates, so its chain law is inherited from that basis; the continuous lifting experiment instead optimizes an unconstrained matrix with a finite compatibility penalty. §10 states plainly that neither validates a categorical claim.

Equality-constrained least squares is classical: Björck [15] surveys constrained least-squares methods, and Eldén [16] develops a direct method for the equality-constrained problem. Lim [17] places graph incidence operators, cycle spaces, and Hodge Laplacians in a unified linear-algebraic treatment; Hoppe and Schaub [18] connect edge flows and higher-order cell-complex structure. These works explain why constraining or shrinking a linear estimator toward a correct cycle subspace can reduce variance. Our contribution is not that mechanism; it is the controlled evaluation and audit trail for deriving the soft penalty from the observed input in this lifting generator.

4. Two settings

4.1 Selection over a fixed family

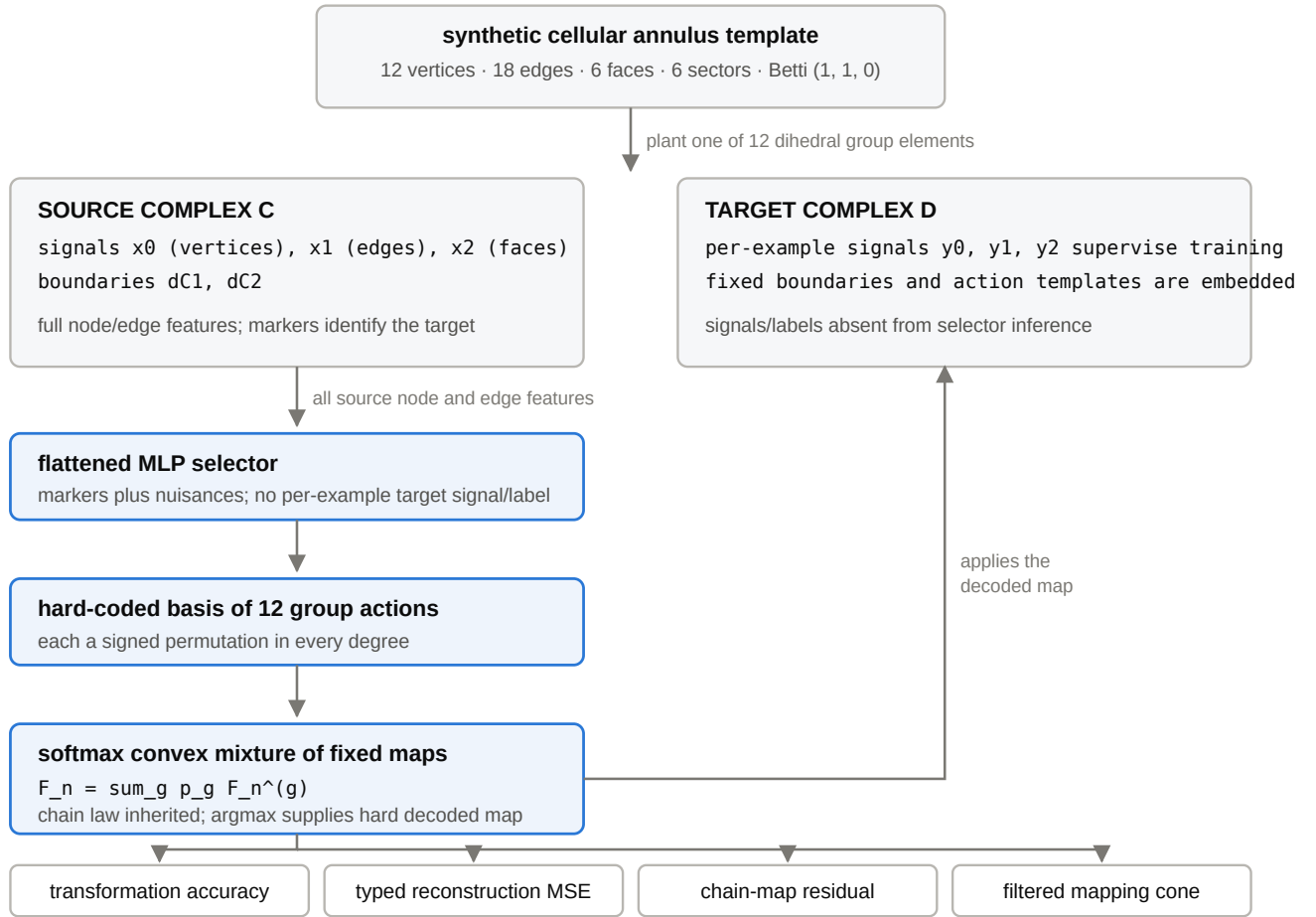


Figure 1. The selection setting. A group element is planted in a six-sector cellular annulus (12 vertices, 18 edges, 6 faces, Betti (1, 1, 0)). The flattened MLP consumes all source node and edge features; identifying markers carry the target identity among nuisance channels. No per-example target signals or labels enter selector inference, but the fixed annulus boundaries and twelve action templates are built into the hypothesis class, and target signals supervise the declared training objectives. During training the model forms a softmax convex mixture of the twelve fixed signed chain maps; hard evaluation uses the argmax vertex. Every mixture satisfies the chain law because every template does, so the residual is a numerical check rather than a learned objective.

Twelve dihedral maps act on the annulus. Each is built as a **signed permutation in every degree**: F_0 permutes vertices, F_1 permutes edges with an orientation sign, F_2 is fixed by matching the mapped boundary to a unique oriented face. We verify numerically that all twelve satisfy $F^T F = I$ at degrees 0, 1, and 2.

The executed `IdentifiableTypedMapModel` encodes the complete source node and edge feature tensors with a flattened MLP, predicts twelve logits, and uses their softmax weights to linearly mix the fixed degree-wise templates. It does not instantiate `ExactChainMapLayer` or a generic nullspace parameterization. The fixed source/target boundaries and all twelve action templates are registered inside the model; only per-example target signals and labels are absent from the selector's forward inputs. An argmax over the same logits supplies the hard decoded transformation used by the discrete accuracy and exact-cone evaluations.

That property determines the exact hard-map certificate in this setting; it does not determine the behavior of soft mixtures during training (§5).

4.2 Learned continuous edge-to-cycle lifting

We use a generator in which the lifting is learnable. Its design is forced by a conflict: the routing benchmark used elsewhere in this project deliberately hides cell structure from the graph observation so a router cannot shortcut, which makes conversion impossible by construction. Routing needs targets *not* inferable from the graph; conversion needs them *inferable*. One generator cannot serve both, so this is a separate one.

- The **2-cells are a cycle basis of the graph**, so the cell complex is determined by the graph rather than drawn beside it. Graph size and density vary, so the cycle rank varies, and the lifting defect varies with it.
- **Face activity thresholds the circulation** of the edge cochain around each cycle — exactly $B_2^\top \times 1$. Integrating an edge feature around a cycle is the operation the lifting must perform.
- **Sheaf transport** combines an endpoint frame difference with a per-edge twist. The frame difference telescopes to the identity around any closed cycle, so without the twist every holonomy would be trivially zero; the twist rides a separate edge channel so holonomy and face activity are not the same quantity twice.

The learning task. For each topology, fit a separate $W: R^E \rightarrow R^F$ from edge signals to cycle-basis coordinates. Its transpose $W^\top: R^F \rightarrow R^E$ is a candidate d_2 ; W is degree-changing and is not a typed chain map between complexes. B_2 is not supplied explicitly to the optimizer, but paired responses generated as $Y = X B_2 + \text{epsilon}$ provide ordinary supervised signal about it. There is no shared model that generalizes W to a new topology. B_1 is observable because it is the graph. Ground truth is $W = B_2^\top$, a median of 242 free parameters across the 29 eligible topologies (range 30–770). The generator uses NetworkX `cycle_basis`, whose basis coordinates are noncanonical. The boundary-compatibility objective identifies the cycle **subspace**, not an arbitrary choice of basis coordinates; paired targets supply the coordinate convention for each fitted topology. The generator enforces $B_1 B_2 = 0$; the degree-1 kernel of the canonical inclusion equals the graph's cycle rank.

This non-explicit provision is not an information-theoretic barrier. With the deterministic generator algorithm known, the NetworkX basis B_2 is algorithmically recoverable from the observed graph, and B_1 alone determines the target cycle subspace $\ker B_1$. The penalty is therefore strong input-derived structural side information, not direct use of B_2 or response labels. The unpenalized baseline ignores B_1 . The campaign omits stronger graph-aware analytic comparators: one could reconstruct the generator's cycle basis directly, parameterize the rows of W in $\ker B_1$, or project an estimator into that subspace with a Hodge/nullspace operator. The reported contrast is against a graph-blind full-matrix baseline, not against an optimal cycle-basis or Hodge oracle.

For each eligible seed, the executed estimation problem is

```
X_train[i,:] ~ N(0, I_E)
Y_train = X_train B2 + epsilon,    epsilon[i,:] ~ N(0, 0.02^2 I_F)
X_test[i,:] ~ N(0, I_E),           Y_test = X_test B2
W_hat_lambda = argmin_W mean((X_train W^T - Y_train)^2) + lambda R(W).
```

The optimizer is Adam for 2,500 steps at learning rate 0.05, initialized at $W = 0$ in float64. The held-out target is noiseless, so its MSE evaluates map recovery. For the distance surrogate, $R(W)$ forms the two 16×16 matrices `cdist(X_train, X_train)` and `cdist(X_train W^T, X_train W^T)`, including their zero diagonals; each matrix is divided by its own full-matrix mean plus $1e-12$, and the elementwise squared difference is averaged.

The sample geometry makes the regularization problem deliberately data-scarce. With $N_{\text{train}} = 16$, the unconstrained row-wise linear system has $E > 16$ in 21 of 29 eligible seeds. For a connected graph, each row in the compatibility-zero set lies in the F -dimensional cycle space; $F \leq 16$ in 24 seeds. Sixteen seeds therefore move from an underdetermined ambient system ($E > 16$) to a potentially identifiable hard cycle-subspace system ($F \leq 16$). Median dimensions are $E = 23$, $F = 11$; five seeds still have $F > 16$. The executed finite penalty does **not** perform that hard dimension reduction, but it shrinks toward the lower-dimensional subspace. A large gain is therefore plausible as scarce-probe system identification, without implying a general advantage in well-sampled regimes.

5. Retrospective no-fit diagnostics for structural objectives

During the post-campaign audit we used two inexpensive checks that require no training. They were not used prospectively to select the reported objectives or predict the reported results. We propose them as pre-fit screens for future structural objectives:

1. **Sample variability:** does the proposed quantity vary over a supplied candidate sample? No detected variation means that sample offers no evidence of discrimination; it does not prove class-wide constancy unless the sample exhausts the class.
2. **Truth-relative ordering:** is the known truth's score no higher than every supplied alternative? If not, the objective favors at least one supplied alternative over the truth. This does not locate a local or class-wide optimum.

These are diagnostic heuristics, not necessary or sufficient conditions for better held-out prediction. In finite samples, even a target-misaligned regularizer can improve generalization through a bias–variance tradeoff; a target-aligned, nonconstant objective can still fail through optimization, scaling, or model misspecification. The claims below therefore rest on the algebraic proof for the annulus and the frozen empirical contrasts, not on the screen alone.

5.1 The hard-map cone certificate is constant on annulus vertices

A mapping cone is acyclic exactly when its chain map is a quasi-isomorphism. Every hard decoded candidate in §4.1 is a signed permutation, hence invertible, hence a quasi-isomorphism. **Exact cone acyclicity therefore takes the same value on all twelve decoded vertices.**

The generalization is broader than this annulus: **any hypothesis class of invertible chain maps between fixed complexes has a constant cone-acyclicity signal.** Acyclicity can certify that a candidate is a quasi-isomorphism; it cannot select among candidates that already have that property.

This statement does **not** extend to the complete training trajectory. Training uses `cone_soft_betti` on convex mixtures of the twelve maps; those mixtures need not be invertible and the differentiable proxy can vary and provide gradients. The RTD-style annulus loss is also not constant: it compares cross-example distances of predicted and target typed representations, the selectors vary by example, and soft mixtures are non-orthogonal. It consumes target batch structure. Accordingly, the empirical chance accuracies in §7.4 cannot be deduced from a constancy theorem for either soft objective.

5.2 The continuous surrogates are target-misaligned

In §4.2 the truth $W = B2^T$ has full row rank. Nevertheless, $\exp(-2 \cdot \sigma_{\min}(W))$ decreases monotonically as the smallest singular value grows, so its optimum is not the finite ground-truth scale: it continually rewards singular-value inflation. The term therefore imposes a directional bias. The frozen experiment establishes that this bias harmed held-out error at the tested weight; the analytic observation alone would not have proved harm.

The RTD-inspired normalized pairwise-distance term has a different mismatch. It asks the source edge vectors X and lifted vectors $X W^T$ to preserve normalized pairwise geometry. The truth $W = B2^T$ is intentionally lossy: because $F < E$, it discards cut-space components and cannot preserve the full source geometry. The surrogate therefore opposes a defining property of the target lifting. Its no-detected-improvement result concerns this target-misaligned surrogate only and says nothing negative about published RTD or SRTD.

5.3 Operational screen

`screen_structural_term` evaluates a candidate term on a known truth and a supplied candidate sample, returning the operational labels `truth-no-higher-than-supplied-candidates-and-varies`, `truth-higher-than-a-supplied-candidate`, or `constant-over-supplied-candidates`. The labels are sample-relative prompts for inspection, not theorems about the full hypothesis class or generalization. The exact annulus class-wide constancy result is instead established analytically; all effect-size and held-out-performance claims come from the reported campaigns.

6. Experimental design

6.1 Prospectively locked same-family replication

The historically named conversion campaign for the edge-to-cycle lifting was frozen in a protocol document committed **before execution**: SHA-256 `503cc282f40d118ba1739c2afe1bfc77eaf2b1733baaddb91c0c3363e75ae2b8`, committed at `d5d18af`, campaign run at `11644c6` from a clean worktree. No endpoint, weight, decision rule, or sample size changed after the freeze. One objective implementation did: the protocol writes the squared Frobenius norm $\|B1 W^T\|^2$, while the runner executed the elementwise mean `mean((B1 W^T)^2)`. Because the number of entries varies by topology, this is not one global rescaling. The paper reports the executed objective and treats the result as a prospectively specified analysis with a disclosed implementation deviation, not a pristine protocol replication or independent confirmation.

The design was locked prospectively only relative to this run. Protocol 27 says that the hypotheses, effect directions, endpoint, training-set size, and tested weights were chosen after exploratory results on the same generator family. Document 26 does not retain the exploratory seed identities, so possible overlap with the frozen seed block 20261001–20261030 cannot be audited. Prospective locking prevents changes after the freeze; it does not remove this outcome-informed design history. Independent confirmation requires an untouched, disjoint generator family or seed block.

item	value
declared topologies	30 (seeds 20261001–20261030)
eligible	29 ; seed 20261025 had fewer than three faces and was skipped, not replaced
training pairs	16
held-out pairs	3072
training-label noise	Gaussian, sigma 0.02
held-out target	noiseless $B2^T x$
primary statistical unit	eligible generator seed: joint topology, data, and noise realization ($n = 29$)
optimiser	Adam, lr 0.05, 2500 steps, W initialised to zeros, float64
weights	exact 3.0, cone 0.01, rtd 0.1

Weights were selected after exploratory work and then frozen before this run. The publication bundle does not retain a machine-verifiable record of every exploratory candidate or the exploratory seeds, so it does not claim that these are globally or even exhaustively “best” weights or that exploratory and frozen seed sets were disjoint. Weights are not comparable across terms because the terms have different scales. The keys `exact`, `cone`, and `rtd` are frozen historical labels defined and qualified in §2.3.

6.2 Endpoints and decisions

Primary endpoint is the paired quantity, one value per eligible generator seed:

$$d = \log_{10}(\text{held-out MSE with term} / \text{held-out MSE without})$$

Negative means the term improves the lifting. Each seed jointly determines the topology, training and evaluation inputs, and training noise; the penalized and unpenalized fits share that realization. Pairing therefore controls the shared realization in each contrast but does not remove topology variance. The estimand is the mean paired log ratio over eligible generator seeds, and the Student-t interval treats the 29 seed-level replicates as exchangeable. Training responses contain Gaussian noise with sigma 0.02, whereas held-out targets are noiseless, so held-out MSE measures recovery of the planted linear lifting rather than prediction of fresh response noise.

Multiplicity is adjusted. The three primary contrasts form one family; the governing interval is Bonferroni-corrected to two-sided 98.333%, giving a family-wise error rate of 0.05. Unadjusted 95% intervals are reported alongside and the adjusted interval governs. An exact two-sided sign test is reported as distribution-free sensitivity and governs nothing.

H4 and H5 were prespecified but are **secondary analyses outside this multiplicity-controlled family**. H4 has an unadjusted Student-t 95% interval; H5 has no valid inferential interval or decision. H4 measures the mean within-seed Pearson correlation, over nine fixed weights for the historically named `exact` objective, between $\log_{10}(\max(\|B1 W^T\|_F, 1e-30))$ and $\log_{10}(\max(\text{held-out MSE}, 1e-300))$. The reported defect is a Frobenius norm; it is distinct from the elementwise mean-square compatibility term used during training. All nine fits within a seed reuse the same generated data and initialization and differ in the common driver, penalty weight `lambda`. They are points on one deterministic regularization path, not nine independent observations.

The frozen H5 endpoint contains a design error discovered during audit. It divides the selected route's error by the per-row minimum of those same two route errors. Consequently every ratio is at least one and every `log10` ratio is nonnegative, while the decision rule required an interval strictly below zero. Support was impossible by construction. We preserve the result for transparency but draw **no inferential routing conclusion** from H5.

The campaign ran once over all declared topologies. No interim analysis, no early stop, no extension.

6.3 Hardware, software, provenance

Campaign: CPU, float64, PyTorch 2.13.0+cu130, Python 3.12.3. The annulus campaign and all timing benchmarks ran on one NVIDIA GB10, Linux 6.17.0-1026-nvidia aarch64, CUDA 13.0, deterministic algorithms enabled with `CUBLAS_WORKSPACE_CONFIG=:4096:8`.

item	value
annulus campaign commit	8021292e97abfec91768f1b5437c883a42c29c60
annulus launch fingerprint	44408d7adf8467e594879b46e25a1cb7fd89a7e7a5d5f3446548bcbf3ed1096e
annulus strict summary SHA-256	0cd0defb0b0d41b5f7563c364cfd6a2cb72c5e5a845bdd5d1ab76a2e1cb953c
routing campaign commit	e69b07707950b6abe332366c51fe8c94254899f3
scheduler steps completed	56 of 56

The sealed scheduler receipt lists all 296 produced files with byte counts and SHA-256 digests; all 296 verify against the on-disk artifacts.

7. Results

7.1 Boundary compatibility improves an edge-to-cycle lifting

objective	weight	95% interval	Bonferroni 98.33%	sign test	decision
boundary compatibility (exact)	3.0	[-2.628, -1.632]	[-2.749, -1.511]	<1e-6	improves
singular-value cone surrogate (cone)	0.01	[+0.125, +0.254]	[+0.109, +0.270]	<1e-6	harms
RTD-inspired distance surrogate (rtd)	0.1	[+0.002, +0.034]	[-0.002, +0.038]	0.458	no detected improvement

Boundary compatibility improves edge-to-cycle lifting

Paired log10 held-out-error ratio; negative favors the objective.
One eligible seed jointly fixes topology, predictors, and training noise.
29 eligible seeds, 16 noisy training probes; same-family replication.
Locked prospectively after outcome-informed weight selection; one execution deviation disclosed.
Thick bar: Bonferroni 98.33% (governs the decision). Thin bar: unadjusted 95%.

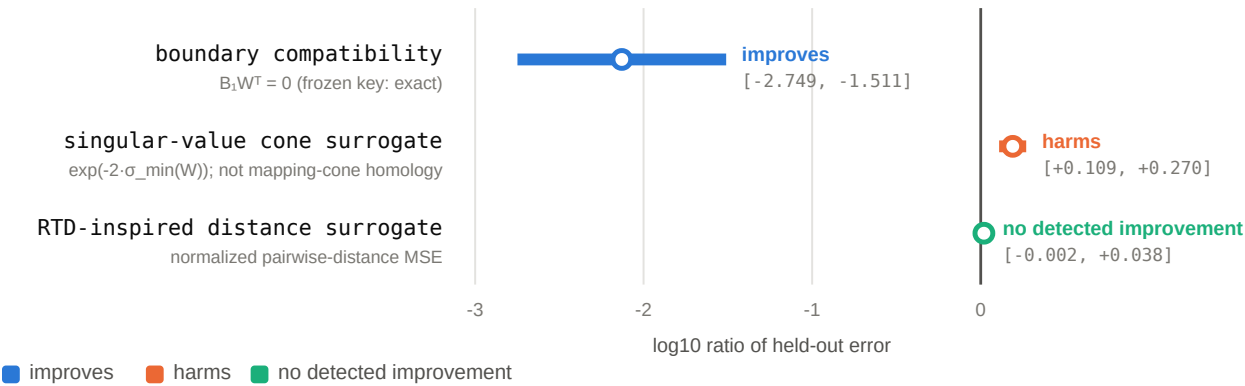


Figure 2. Three structural objectives under one protocol. Paired log10 (held-out error with term / without), one value per eligible generator seed, 29 joint topology/data/noise realizations. The thick bar is the Bonferroni-adjusted 98.33% interval governing the primary decision; the thin bar is the unadjusted 95% interval. Boundary compatibility improves held-out map recovery; the singular-value surrogate harms it; the RTD-inspired distance surrogate shows no detected improvement. The short labels in the graphic reproduce frozen campaign keys and do not identify exact with sequence exactness, cone with mapping-cone homology, or rtd with published RTD. Generated from results/campaigns/conversion-campaign-v1-corrected.json.

Median `log10` ratio for boundary compatibility is -2.858 , roughly a **700-fold** reduction in held-out error; the mean is -2.130 . The adjusted interval lies far from zero.

The executed term is `mean((W B1T)2)`, which is zero at the truth because $B2^T B1^T = (B1 B2)^T = 0$. It does not directly use `B2` or response labels, but it uses `B1`, which determines the target cycle subspace; the unpenalized baseline ignores this strong structural side information. With the generator algorithm known, `B2` is algorithmically recoverable from the graph, and no analytic cycle-basis, nullspace, or Hodge-projection oracle was included. Training still optimizes all `F×E` entries of `W`; weight 3.0 penalizes but does not prohibit off-cycle components.

7.2 Secondary analysis: covariation along the compatibility-penalty path

Within each topology, nine prespecified weights on the compatibility penalty produce learned maps of varying quality. The endpoint is the correlation between a map's boundary-compatibility defect and its held-out error along that path.

quantity	value
mean within-seed correlation	+0.961
95% interval	[+0.935, +0.987]
eligible seeds with positive correlation	29 / 29

This prespecified analysis is outside the three-contrast multiplicity family, so its 95% interval is unadjusted. The same data and initialization are reused for the nine fits, and `lambda` is a common driver of both compatibility defect and held-out error. The nine path points are therefore not independent observations. The result establishes covariation along this regularization path; it does not show that defect carries independent predictive information, calibrates error off path or on unseen topologies, isolates an intervention effect, or is causal.

7.3 The singular-value surrogate harms; the distance surrogate does not improve

Both belong to the multiplicity-controlled primary family. The singular-value cone surrogate harms held-out error at the tested weight: its adjusted interval lies entirely above zero. For the RTD-inspired normalized pairwise-distance surrogate, the adjusted interval contains zero and the sign test is 0.458. This supports **no detected improvement**, not equivalence, inertness, or a general claim about published RTD/SRTD; no equivalence margin was prespecified.

§5.2 identifies the singular-value surrogate's scale-inflating bias; the frozen contrast establishes that it harmed here. §5.1 proves a different fact in the selection setting: exact cone acyclicity is constant across that finite class.

7.4 The selection setting, and why its nulls are weak

On the annulus, all six objectives containing task or reconstruction supervision decode the planted transformation perfectly: transformation and cell-face accuracy are 1.000 on all five seeds. Their mean map MSE spans $2.618e-17$ to $2.504e-8$. The engineering recovery gate applies only to `task_reconstruction` and `combined` — five seeds each — and passes **10 of 10** applicable runs.

Recovery by training objective

Mean over five seeds. Objectives without task/reconstruction sit at chance.

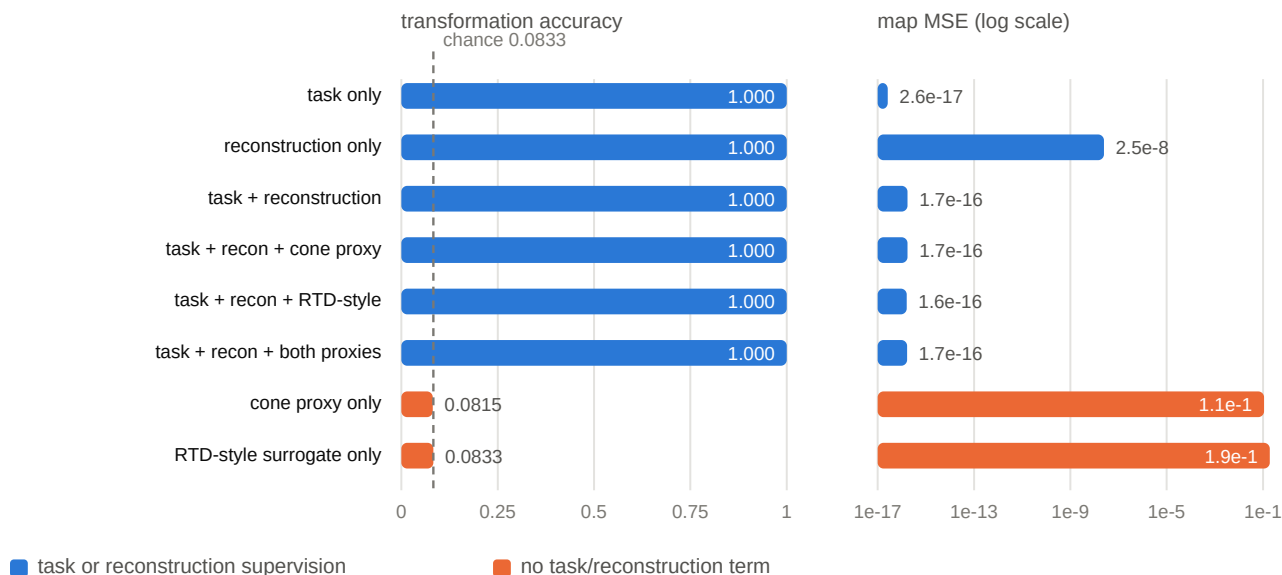


Figure 3. Recovery by objective in the selection setting. Mean over five seeds. All six objectives carrying task or reconstruction supervision have perfect decoded accuracy, with mean map MSE from $2.618\text{e-}17$ to $2.504\text{e-}8$. The cone-only and RTD-only controls sit at chance with mean map MSE 0.109 and 0.191, about 7–16 orders of magnitude above the supervised range depending on the comparison. The RTD-only loss consumes target batch geometry and is not an unsupervised control. Generated from `results/summaries/identifiable-campaign-summary.json`.

All 21 declared continuous contrast intervals contain zero, so adding the annulus cone proxy or RTD-style training surrogate changed nothing. **That null is weak evidence:** an analytic marker decoder also reaches 1.000, so the ceiling was attainable without learning and no candidate could improve on a control already at the maximum.

The two single-loss controls are descriptive negative results. `cone_only` reaches transformation accuracy 0.0815 and `rtd_only` 0.0833 against a 0.0833 chance baseline — **while producing acyclic cones in 6,000 of 6,000 evaluated examples each**. This shows that the exact hard-map certificate is satisfied by all decoded predictions while identification remains at chance. It does not show that the differentiable cone proxy was constant during training, and it supplies no analogous constancy proof for the RTD-style loss.

7.5 Specificity against generic regularisation remains open

Earlier exploratory notes report ridge, random-subspace, and nonlinear-link checks, but no generating script or machine-readable result for those analyses is preserved in the publication evidence. Their quantitative values are therefore excluded from this paper. The primary campaign establishes an improvement over the unpenalized fit; it does not establish that boundary compatibility outperforms a tuned generic regularizer or a rank-matched random subspace. A frozen, machine-recorded matched-control campaign is required before making that specificity claim.

7.6 Routing: a flawed accuracy endpoint and a separate compute result

The prespecified H5 accuracy endpoint is **non-informative because its decision rule is impossible to satisfy**. It divides the chosen route's error by the per-row minimum of the cell and graph errors, so every ratio is at least one and every log ratio is nonnegative. An interval strictly below zero cannot occur.

The original summary had a second flaw: its 28 rows are two tested weights for each of only 14 topology clusters. Treating them as 28 independent observations and using a Student-t interval with 27 degrees of freedom is pseudoreplication. The formerly reported interval and 25-of-28 selection count are withdrawn from inferential use. The corrected record reports the historical row-naïve interval $[-0.111, +0.382]$ and a topology-clustered descriptive interval $[-0.123, +0.393]$; neither is inferential, and H5 receives no decision. A redesigned, cluster-aware endpoint is required.

A separate routing result, from a different experiment, is retained as a descriptive compute comparison rather than an accuracy claim.

Trained routing inference latency on GB10

Median over 100 timed iterations, averaged across five seeds; whisker marks mean p95.
Batch 64, bfloat16. Paths were timed in this order inside one process.

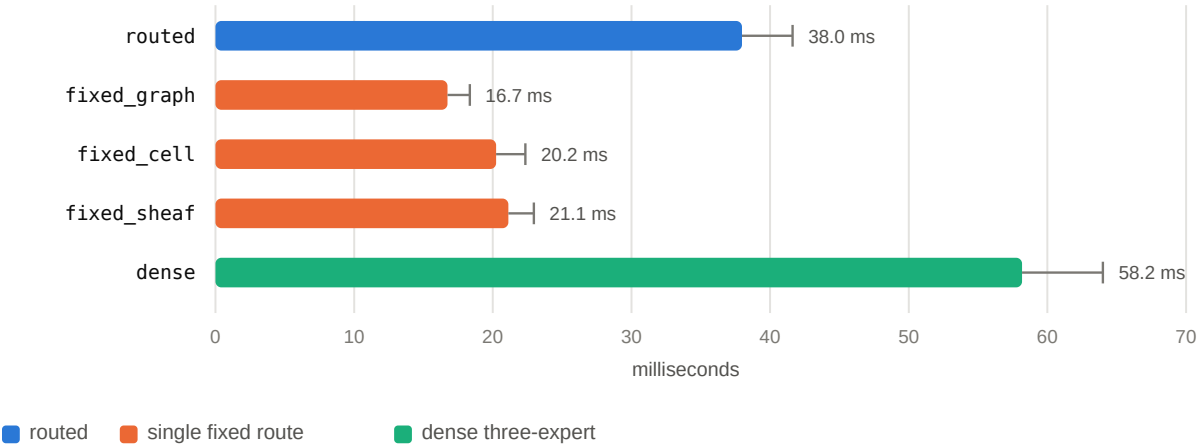


Figure 4. Trained routing inference latency on GB10. Median over 100 timed iterations averaged across five seeds; whiskers mark mean p95. Batch 64, bfloat16. All paths were timed in the plotted order inside one process, so residual thermal or allocator drift is confounded with path order. Generated from `results/summaries/compute-campaign.json`.

Across five seeds, the mean dense/routed speed ratio is **1.532** (Student-t 95% CI [**1.489, 1.575**]), and the mean routed/fastest-fixed latency ratio is **2.269** (95% CI [**2.215, 2.322**]). Routed peak allocated memory is below dense in every seed. The defensible statement is that routing saves compute against dense evaluation, not that it saves compute overall and not that a measured defect selects the view.

An earlier five-seed campaign descriptively found a hard-minus-best-fixed accuracy margin of +0.1098, 95% interval [+0.0953, +0.1243]. That result used privileged latent-regime distillation and had target structured views available at inference, and a disclosed pre-freeze procedural deviation makes it protocol-aligned rather than pristine preregistration. It is a different experiment from the defect-based routing tested here and the two must not be conflated.

7.7 Corruption diagnostics

Every corruption contrast interval contains zero

Candidate minus baseline on the adjusted partial-Spearman endpoint, with 95% intervals.
Gate-3 rows are conditional on a fixed checkpoint pair; gauge rows are across eight training seeds.

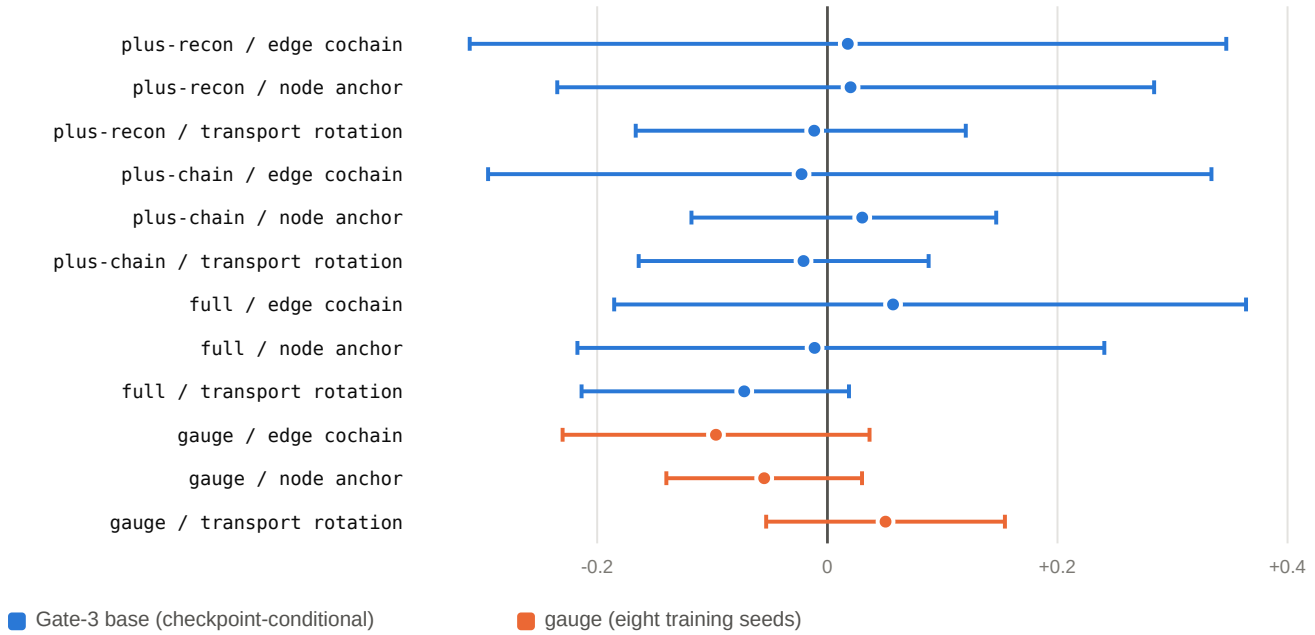


Figure 5. Every corruption contrast interval contains zero. Nine Gate-3 base contrasts use a paired complete-block bootstrap conditional on a fixed checkpoint pair; three gauge contrasts use a Student-t interval with $df = 7$ across eight training seeds. No multiplicity adjustment. Generated from `results/gate3/paired_comparison_final.json` and `results/summaries/gauge-corruption-campaign.json`.

These compare clean and corrupted **fixed-expert embeddings**. They never invoke a translator or learned map and therefore cannot support or refute any conversion claim. All twelve intervals contain zero.

8. Discussion

The precise objective matters. In one prospectively locked same-family replication with a disclosed implementation deviation, boundary compatibility improves held-out recovery of an edge-to-cycle lifting, a singular-value cone surrogate harms it, and an RTD-inspired normalized pairwise-distance surrogate shows no detected improvement. Reporting only that "homological structure helps" or "does not help" would erase the distinction between a boundary-of-boundary penalty and two motivated but nonidentical surrogates.

Retrospective no-fit inspection is useful but not dispositive. §5's checks expose the hard-map cone constancy result and the singular-value surrogate's scale-inflating bias. They guide experimental design; they neither predict an effect size nor replace held-out evaluation.

Acyclicity is not correctness. An acyclic mapping cone certifies that a chain map is a quasi-isomorphism. In the annulus class every candidate is already an invertible chain map, so that certificate is constant; a model trained on the cone proxy alone can satisfy the certificate while failing to identify the planted map.

The cone bundles what the kernel and cokernel separate. $\dim H_n(\text{cone}) = \dim \text{coker } H_n + \dim \ker H_{(n-1)}$ — the cone sums a directional pair and loses the distinction between *destroyed* and *unreachable*. Where a measurement is wanted, the unbundled pair is strictly more informative.

The boundary-compatibility penalty favors the correct input-derived subspace. Its zero set has every row of W in the cycle space, which has dimension F for these connected topologies. The executed finite-weight regularizer merely shrinks off-cycle components; it does not hard-constrain the optimizer, reduce the parameter count from $F \times E$, or prove exactness. The result shows that this cycle-subspace bias helped the generated targets and was available from the observed graph. This is strong side information: the unpenalized comparator ignores $B1$, while a generator-aware cycle-basis oracle or Hodge/nullspace estimator could use it even more directly. Whether the gain persists against those graph-aware comparators or a tuned generic regularizer remains open.

The effect occurs in a favorable scarce-probe geometry. With only 16 probes, 21 of 29 ambient systems have $E > 16$; the hard cycle-subspace dimension is at most 16 in 24 seeds, and 16 seeds cross from $E > 16$ to $F \leq 16$. The finite penalty does not literally reduce the fitted $F \times E$ parameter count, but its shrinkage targets the subspace that a hard graph-aware estimator would use. The large observed effect is therefore consistent with classical variance reduction in underdetermined system identification.

9. Limitations

- **One synthetic linear lifting family.** 29 eligible generator seeds and one training-set size. Nothing transfers automatically to real data or nonlinear transformations.
- **Seed-level inference.** Each primary replicate jointly instantiates topology, data, and training noise. Intervals assume the 29 eligible seed-level paired effects are exchangeable; pairing controls their shared realizations but does not remove topology variance.
- **No unseen-topology learning.** A separate W is fitted and evaluated within each topology. The campaign does not learn a shared converter that generalizes to a new graph.
- **Noncanonical target coordinates.** NetworkX chooses one cycle basis among many. The penalty identifies the cycle subspace; paired labels identify that run's arbitrary basis coordinates.
- **Strong structural side information and a missing oracle.** Although the penalty does not directly use $B2$ or response labels, $B1$ determines the target cycle subspace, and the known deterministic generator makes $B2$ algorithmically recoverable from the graph. The unpenalized baseline ignores $B1$; no analytic cycle-basis, Hodge-projection, or nullspace baseline was run.
- **Favorable scarce-probe geometry.** $N_{\text{train}} = 16$; 21/29 seeds have $E > 16$, 24/29 have $F \leq 16$, and 16/29 cross from the former ambient underdetermination to the latter hard-subspace dimension. Medians are $E = 23$, $F = 11$, while five seeds still have $F > 16$. The executed finite penalty is shrinkage, not a hard dimension reduction, and the effect may diminish with more probes.
- **Protocol implementation deviation.** The runner used an elementwise mean where frozen protocol 27 wrote a Frobenius sum. The result is not a pristine protocol replication.
- **Regularization specificity is open.** No machine-verifiable tuned-ridge or rank-matched random-subspace campaign is in the publication evidence.
- **Outcome-informed same-family design.** The hypotheses, directions, endpoint, training size, and weights were selected after exploration on the same generator family. Exploratory seed identities were not retained, so overlap with the frozen 20261001–20261030 block is unverifiable. This is prospectively locked replication, not independent confirmation; an untouched disjoint family or seed block remains necessary.

- **The selection setting saturates.** Six of eight objectives reach exactly 1.000, so its structural nulls cannot detect improvement. Its RTD-only loss consumes target batch geometry and is not an unsupervised control.
- **Five seeds in the annulus campaign**, where the exact two-sided sign test floor is $p = 0.0625$ and can never be decisive.
- **Defect-based routing accuracy is unresolved.** The frozen H5 decision was impossible to satisfy because its per-row oracle denominator forces every endpoint observation to be nonnegative. Its 28 rows also comprise only 14 topology clusters at two weights, so the former $df=27$ interval and 25/28 count cannot be treated as independent-trial inference. H5 receives no decision.
- **Privileged supervision** in the historical routing campaign, and **target-view translators** in the historical conversion modules, which consumed target structure and are not conversions.
- **Timing caveats.** All paths timed in fixed order inside one process; raw per-iteration timings not retained; identifiable runner reports p90 and routing runner p95, never pooled.
- **Secondary analyses are unadjusted.** C1, the separate routing campaign, and the corruption diagnostics lie outside the three primary lifting contrasts; H5 has no valid inferential decision.
- **C1 is a regularization-path association.** Within each eligible seed, its nine fits share data and initialization and vary the common driver `lambda`. They are not independent observations, so C1 does not establish independent predictive information or off-path calibration.
- **No equivalence test.** The RTD-inspired surrogate interval includes zero, but no equivalence margin was specified.
- **The distance surrogate is target-misaligned.** It asks a rank-`F` lifting to preserve full source geometry even though the truth intentionally discards cut-space components. Its no-detected-improvement result does not test or criticize published RTD/SRTD.
- **Artifact boundaries.** The 8.8 GB `artifacts/` tree is untracked; tracked evidence is the curated `results/` bundle.

10. Claim boundary

This work provides evidence, on one synthetic family under a prospectively locked same-family replication with a disclosed implementation deviation, that an input-derived boundary-compatibility penalty improves an edge-to-cycle lifting relative to an unpenalized full-matrix baseline that ignores `B1`. The hypotheses, directions, endpoint, training size, and weights were outcome-informed by same-family exploration, and exploratory seed overlap is unverifiable; this is not independent confirmation. A prespecified, unadjusted secondary analysis finds that defect covaries with held-out error along the fixed penalty path, where `lambda` is a common driver. In the same primary family, a singular-value cone surrogate harms held-out performance, while an RTD-inspired normalized pairwise-distance surrogate shows no detected improvement.

It does **not** establish:

- any general conclusion about mapping-cone homology or published RTD/SRTD as training objectives;
- that a measured defect can select a representation; the frozen test of this claim had an impossible decision rule;
- lifting quality on real or out-of-distribution data, or any broader conversion claim;
- superiority to an analytic cycle-basis, Hodge-projection, or nullspace method;
- independent replication on an untouched generator family or disjoint seed block;
- independent predictive information or off-path calibration from the C1 regularization-path association;
- general equivalence between graphs, cellular complexes, and sheaves;
- a general method for learning quasi-isomorphisms — the annulus candidates are preconstructed invertible chain maps, and the verified chain-map identity uses a fixed $1e-5$ tolerance;

- any Langlands, eigensheaf, Fourier–Mukai, or category-theoretic result. Imposing a chain-map constraint by construction is not a categorical claim in the sense of [9] or [10]: we fix one finite hypothesis class by hand in §4.1, while §4.2 optimizes a full matrix under a finite soft penalty rather than learning inside a fixed nullspace.

11. Code and data availability

Availability status. The project repository is private and is licensed proprietarily (Copyright © 2026 Sean Mahdavian, all rights reserved). This is **not an open-source or open-data release**.

Access for peer review. The source, configurations, frozen protocols, and compact evidence bundle are made available to editors and reviewers for the duration of review through a read-only snapshot supplied to the handling editor. The snapshot is built by `scripts/build_review_snapshot.py`, which archives the repository at one commit together with the tracked `results/` bundle and its manifest and refuses to run against an uncommitted worktree. Reviewers can rehash the compact evidence, trace every reported value to it, rerun the CPU edge-to-cycle lifting campaign, run the test suite, and regenerate the figures and paper. The snapshot intentionally excludes the 8.8 GB raw `artifacts/` tree. A full rerun of the GB10 annulus campaign or trained-checkpoint benchmarks therefore requires separately supplied raw artifacts/checkpoints and compatible GPU hardware. Access is granted for review only and confers no license to redistribute or reuse the material; the terms in `LICENSE` continue to apply.

Readers outside the review process should treat this section as a description of what exists and how it is organised, not as a download.

All code, configurations, and compact evidence are in that repository. The tracked bundle under `results/` contains 50 files with a `MANIFEST.json` recording path, byte count, SHA-256, generating commit, and generating command for each.

location	contents
<code>results/campaigns/</code>	the frozen edge-to-cycle lifting campaign record
<code>results/summaries/</code>	strict compact summaries for the annulus, gauge, compute, and routing campaigns
<code>results/gate3/</code> , <code>results/gate3g/</code>	gate decisions and per-batch corruption-report derivatives
<code>results/benchmarks/</code>	ten identifiable and five routing trained benchmark records

Corruption reports are exported as per-batch derivatives: the `per_example` array is dropped and the `per_batch` array — the unit of analysis — retained. This is lossless for every published statistic, verified by recomputing the adjusted partial Spearman from retained rows and matching to 1e-15.

Synthetic generators are deterministic given the committed configs and seeds; no external dataset is required.

12. Reproducibility

The reviewer snapshot supports the following checks without a GPU:

```
uv sync --frozen --extra dev --python 3.12.3
env CUDA_VISIBLE_DEVICES=-1 .venv/bin/python \
  scripts/run_conversion_campaign.py \
  --output /tmp/conversion-campaign.json
.venv/bin/python scripts/export_publication_evidence.py --verify-only
.venv/bin/python scripts/render_figures.py
.venv/bin/python scripts/render_paper.py
```

The campaign runner fails before fitting unless the lockfile, Python, NetworkX, NumPy, base Torch version, generator, and frozen protocol match the recorded environment. Hiding CUDA keeps this float64 CPU run independent of accelerator occupancy.

With separately supplied raw campaign artifacts and checkpoints, the full private workspace additionally supports:

```
python scripts/summarize_gauge_corruption_campaign.py \
  --output results/summaries/gauge-corruption-campaign.json
python scripts/summarize_compute_campaign.py \
  --output results/summaries/compute-campaign.json
```

Each summarizer revalidates provenance before aggregating and fails closed on any hash, seed, pairing, schema, or receipt mismatch. The exporter works from an explicit allowlist and refuses checkpoints, prediction dumps, histories, logs, and caches. Its manifest carries no timestamp, so re-exporting unchanged evidence reproduces the bundle byte for byte.

Thus the compact snapshot supports verification of every reported value and a fresh CPU conversion run. It does not, by itself, support a fresh GB10 campaign or checkpoint benchmark; those require the excluded raw artifacts and hardware described in §11.

The paper PDF is rendered with `python scripts/render_paper.py`; figures are regenerated from tracked evidence with `python scripts/render_figures.py`.

13. Declarations

Author contributions. Sean Mahdavian designed and implemented the system, ran the campaigns, and wrote the manuscript.

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Competing interests. The author declares no competing interests.

License. The software, source code, documentation, and evidence described here are made available under a proprietary license, Copyright © 2026 Sean Mahdavian, all rights reserved. The terms are in the repository's `LICENSE`. **This is not an open-source or open-data release**, and §11 should be read accordingly.

Ethics. Every experiment reported here uses synthetic data generated by committed deterministic code. No human subjects, personal data, or personally identifiable information are involved. Compute is small: the annulus campaign is 25.8 minutes of GPU time and the edge-to-cycle lifting campaign runs on CPU.

A molecular transfer experiment on the public OGBG-MOLHIV benchmark was run earlier in this project and is **deliberately not reported here**, because it answers a different question from the one this paper asks. Its result — the ring-lift route losing to the graph route, with later variants contaminated by repeated inspection of the official test split — is recorded in the project's claims ledger as C6.

Reporting. Null and negative results are reported in the body rather than an appendix, and every quantitative experimental result is traceable to a tracked machine-readable file under `results/`.

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